

Indian Institute of Technology Patna

Date: 26th February 2025

Midsem CS5203/563: Natural Language Processing

Duration: 2 hours

Full Marks = 50

Instructions:

- All sections should be answered in the main copy provided (not in the question paper).
- There is no negative marking
- No doubts will be entertained.

Section-I

Full Marks: 25

- Given the text: "The quick brown fox jumps over the lazy dog", what would be the output of the regex `\b\w{5}\b`? [1 mark]
 - quick, brown, jumps
 - The, fox, over, dog
 - jumps, quick
 - quick, brown, jumps, over
- What is the primary purpose of Teacher Forcing in training Recurrent Neural Networks (RNNs) for NLP tasks? [1 mark]
 - To prevent overfitting by randomly masking input tokens.
 - To speed up training by using the true previous token instead of the predicted one.
 - To make the model completely independent of past predictions.
 - To replace the need for sequence-to-sequence models in NLP.
- What is the main advantage of Weight Tying in neural language models? [1 mark]
 - It reduces the number of trainable parameters, making the model more efficient.
 - It improves the model's ability to process long-range dependencies.
 - It prevents the vanishing gradient problem in deep networks.
 - It eliminates the need for embeddings in NLP models.
- Which of the following pretrained Transformer models is NOT typically used for text-based tasks? [1 mark]
 - BERT
 - GPT-3
 - ResNet
 - LLAMA
- What is the minimum number of operations (insertions, deletions, substitutions) required to convert "horse" into "ros" using the Levenshtein distance algorithm? [1 mark]
 - 2
 - 3
 - 4
 - 5
- A corpus has 10,000 unique words in its vocabulary. If we use the Bag-of-Words (BoW) model, what is the length of the vector representation of each sentence? [1 mark]
 - Sentence length
 - 10,000
 - Varies based on sentence length
 - 5,000
- A Transformer model computes attention scores for 3 words in a sequence. Given the unnormalized attention scores: [5, 2, 3]. Applying the softmax function, what is the probability of the first word? [1 mark]
 - 0.659
 - 0.244
 - 0.097
 - 0.412
- A machine translation system produces:
Candidate: "A dog is running outside." Reference: "A dog runs outdoors." The 1-gram matches are: "A", "dog"
The candidate sentence has 5 tokens. What is the 1-gram precision? [1 mark]
 - 0.4
 - 0.5
 - 0.6
 - 0.7
- What is the main limitation of an N-gram model? [1 mark]
 - Requires large labeled data
 - Cannot handle out of vocabulary words
 - Ignores syntax
 - All of the above
- In word embeddings, which of the following methods is NOT commonly used for learning word representations? [1 mark]
 - Word2Vec
 - GloVe
 - TF-IDF
 - BERT
- Which of the following is NOT an advantage of using word embeddings over one-hot encoding? [1 mark]
 - Word embeddings capture semantic similarity
 - Word embeddings are high dimensional and sparse compared to one-hot encoding

- (c) Word embeddings enable better generalization in NLP tasks
- (d) Word embeddings reduce dimensionality compared to one-hot encoding

12. A spam classifier based on Naïve Bayes calculates the probability of an email being spam (S) or not spam ($\neg S$) given the presence of the word "offer".

Given:

- $P(S) = 0.4$
- $P(\neg S) = 0.6$
- $P(\text{"offer"} | S) = 0.7$
- $P(\text{"offer"} | \neg S) = 0.3$

Using Bayes' theorem, what is $P(S | \text{"offer"})$? [2 mark]

- (a) 0.44
- (b) 0.60
- (c) 0.58
- (d) 0.64

13. Consider a Transformer-based model processing a sentence of length 10 with a hidden size of 64. The attention scores are computed using the formula:

$$A = \frac{QK^T}{\sqrt{d}}$$

where Q (queries) and K (keys) have dimensions (10×64) , and $d = 64$. How many scalar attention scores are computed in the attention matrix? [2 marks]

- (a) 200
- (b) 100
- (c) 640
- (d) 32

14. Let's say an email could be classified into 3 classes "Urgent", "Normal" and "Spam". A classifier is used to predict these classes. Following is the confusion matrix.

Gold Label (column)			
Predicted (row)	Urgent	Normal	Spam
Urgent	8	10	1
Normal	5	60	50
Spam	3	30	200

Answer Following questions from this confusion matrix. [5x2=10 marks]

- (a) Find out precision and recall for "Urgent" mail.
- (b) Find out precision and recall for "Normal" mail.
- (c) Find out precision and recall for "Spam" mail.
- (d) Find out microaverage precision for classifier
- (e) Find out macroaverage precision for classifier.

Section-II

Full Marks: 15

Attempt any 3 questions

1. Through a step-by-step process, calculate TF-IDF for the given corpus and mention the word(s) having the highest value. (5 marks)

Document 1: We are going to Mumbai

Document 2: Mumbai is a famous place.

Document 3: We are going to a famous place.

Document 4: I am famous in Mumbai.

2. You are tasked with building a machine learning model to classify movie reviews as either positive or negative. You are provided with a dataset of 1,000 reviews, each containing a sentence or a short paragraph. The reviews are preprocessed with tokenization, and the words are transformed into vectors using the TF-IDF method.

(a) For the given pair of movie reviews, the TF-IDF vectors are as follows:

Review 1 (Vector): [0.2, 0.5, 0.3, 0.0, 0.1]

Review 2 (Vector): [0.1, 0.4, 0.3, 0.3, 0.2]

Calculate the cosine similarity between the two reviews. Show your steps and interpret the similarity value. [2 marks]

(b) You now train a logistic regression model on the 1,000 movie reviews, with the target variable being the sentiment label (positive or negative). The model uses the cosine similarity of the reviews (computed during pre-processing) as an additional feature alongside the TF-IDF vector representation. The logistic regression coefficients learned after training the model are:

i. Intercept (Bias): -1.2

ii. Cosine Similarity Coefficient: 3.5

iii. TF-IDF Feature Coefficients: [0.5, -0.3, 1.2, 0.7, -0.2]

You are given the cosine similarity between a new review and a known positive review as 0.85, and the new review's TF-IDF vector is [0.3, 0.1, 0.2, 0.0, 0.5]. Using the logistic regression formula and considering cosine similarity and its coefficient as one of the weight and bias terms, predict the sentiment of the new review (positive or negative). Show the steps for the calculation. [3 marks]

3. Compare RNN, LSTM, and Transformer models in the context of Natural Language Processing (NLP) [compare in tabular manner]. [5 marks]

4. Explain self attention mechanism in transformer with block diagram. [5 marks]

Section-III

Full Marks: 10

Attempt any one question

1. (a) Write the algorithm for Min Edit Distance. Note: Assume insertion and deletion cost is 1 while substitution cost is 2. [4 marks]

N	9									
O	8									
I	7									
T	6									
N	5									
E	4									
T	3									
N	2									
I	1									
#	0	1	2	3	4	5	6	7	8	9
#	E	X	E	C	U	T	I	O	N	

Table 1:

I	want	to	eat	chinese	food	lunch	spend
2533	927	2417	746	158	1093	341	278

Table 3: Showing unigram count

- (d) Explain the problem with zero probability bigram and possible remedy. [2 marks]

- (b) Use the algorithm to fill up Table 1 (3 marks)

- (c) Perform a backtrace and depict how the two strings align. (3 marks)

2. (a) A company has implemented a chatbot that helps customers with their queries. The chatbot uses an N-gram language model trained on previous customer interactions, FAQs, and support tickets. The goal is to predict the next words in a conversation and improve response accuracy. Assume that we have the following word counts and system is using a **trigram language model**.

"I want a" → 400 times
 "I want to" → 300 times
 "I want my" → 200 times
 "I want a refund" → 250 times

If a customer types "I want", what is the probability that next word would be "a"? [2 marks]

- (b) Table 2 and 3 are showing bigram and unigram frequency of few words. Assume $p(I|s) = 0.25$ and $p(/s|food) = 0.68$, find out the probability of sentence $\langle s \rangle I \text{ want chinese food} / s \rangle$. Note: use bigram language model. [4 marks]

	I	want	to	eat	chinese	food	lunch	spend
I	5	827	0	9	0	0	0	2
want	2	0	608	1	6	6	5	1
to	2	0	4	686	2	0	6	211
eat	0	0	2	0	16	2	42	0
chinese	1	0	0	0	0	82	1	0
food	15	0	15	0	1	4	0	0
lunch	2	0	0	0	0	1	0	0
spend	1	0	1	0	0	0	0	0

Table 2: Showing raw bigram count, for example "want" appears 827 times after "I"

- (c) Find out the perplexity of $\langle s \rangle I \text{ want chinese food} / s \rangle$. [2 marks]

